The correct answer for each question is indicated by a ✓.

1 INCORRECT

Which of the following is true of the purple non-sulfur bacteria?

A) They oxidize elemental sulfur to sulfate

B) They perform oxygenic photosynthesis

✓ **C)** Usually are photoorganoheterotrophs

D) Utilize bacteriochlorophyll *c* and bacteriorhodopsin

2 INCORRECT

Purple non-sulfur bacteria are usually found

A) In soil rich in organic matter

✓ **B)** In the mud and water of lakes and bonds with organic matter and low sulfide levels

C) In mammal hosts

D) In eutrophic estuaries

3 INCORRECT

Photoheterotrophs that metabolize carbon and other organic materials using oxygen-dependent respiration are capable of

A) Oxygenic photosynthesis

B) Anoxygenic photosynthesis

C) Fermentation

✓ **D)** Aerobic anoxygenic photosynthesis (AAnP)

4 INCORRECT

Rhodospirillum rubrum, a purple non-sulfur bacterium, exhibits which type of metabolism?

A) Oxydative phosphorylation

B) Fermentation

C) Photoheterotrophy

✓ **D)** All of the above

5 CORRECT

Which of the following is true of the genus *Rickettsia*?

✓ **A)** They are all parasitic or mutualistic

B) Primarily use glycolysis to obtain energy

C) They are evolutionarily similar to chloroplasts

D) All of the above

6 INCORRECT

The families *Caulobacteraceae* and *Hyphomicrobiaceae* both exhibit which of the distinguishing feature?

A) Heterocysts when in nitrogen limited conditions

✔ **B)** A life cycle that features a prostheca with reproduction by budding

C) Bacteriochlorophylls *a* and *c* and carotenoid pigments

D) All of the above

7 INCORRECT

Which of the following describes the function of prosthecae in the genus *Caulobacter*?

A) Gives the cell extra buoyancy

B) Increases the surface area of the cell allowing for nutrient uptake in oligotrophic environments

C) Aids in reproduction by budding

✔ **D)** All of the above

8 INCORRECT

Which group of bacteria form symbiotic relationships with plant roots and perform nitrogen fixation?

A) Purple non-sulfur bacteria

✔ **B)** *Rhizobales*

C) *Caulobacteraceae*

D) *Burkholderiales*

9 CORRECT

Nitrifying bacteria and archaea are capable of

✔ **A)** Oxidizing ammonia to nitrate or nitrate to nitrite

B) Creating nitrogen gas from ammonia

C) Reducing nitrite to ammonia

D) All of the above

10
INCORRECT

The group of bacteria that are capable of utilizing a wide variety of organic molecules in anoxic habitats are the

A) Alpha-proteobacteria

B) Beta-proteobacteria

C) Gamma-proteobacteria

D) Delta-proteobacteria

11
CORRECT

The causative agent of gonorrhea is a member of which order?

A) *Neisseriales*

B) *Burkholderiales*

C) *Nitrosomonadales*

D) *Hydrogenophilales*

12
INCORRECT

Which of the following is true of the order *Burkholderiales*?

A) Most are motile with a single polar flagellum or tuft of polar flagella

B) Includes microbes that are capable of degrading a variety of organic molecules

C) Includes sheathed microbes

D) All of the above

13
INCORRECT

The genus *Thiobacillus*, a member of the order *Hydrogenophilales*, exhibits which type of metabolic activity?

A) Methylotrophy

B) Oxidizing hydrogen sulfide to sulfur

C) Oxidizing a variety of inorganic sulfur compounds to sulfate

D) All of the above

14
INCORRECT

Members of this order are responsible for the corrosion of concrete and pipe structures through the production of sulfuric acid.

A) *Neisseriales*

B) *Burkholderiales*

C) *Nitrosomonadales*

D) *Hydrogenophilales*

15 INCORRECT

Which subgroup of proteobacteria is the largest with extensive diversity?

A) Alpha-proteobacteria

B) Beta-proteobacteria

C) Gamma-proteobacteria

D) Delta-proteobacteria

16 INCORRECT

The purple sulfur bacteria are part of which bacterial supergroup?

A) Alpha-proteobacteria

B) Beta-proteobacteria

C) Gamma-proteobacteria

D) Delta-proteobacteria

17 INCORRECT

Microbes that oxidize and form microbial mats on the nutrient-rich seafloor and at deep hydrothermal vents are members of the order

A) *Enterobacteriales*

B) *Thiotrichales*

C) *Pseudmonadales*

D) *Vibrionales*

18 INCORRECT

Which of the following is true of *Thiomicrospira crunogen* XCL-2, a deep-sea microbe isolated from a hydrothermal vent?

A) Has a high affinity for HCO_3^- and CO_2 under low CO_2 conditions

B) Contains a multienzyme complex capable of reducing sulfur compounds to sulfate

C) Includes a complete prophage within its chromosome

D) All of the above

19 INCORRECT

Methylotrophy is observed in which of the following orders?

A) *Thiotrichales*

B) *Methylococcales*

C) *Pseudomonadales*

D) *Alteromonadales*

20 CORRECT

Which of the following is true of bacteria in the order *Legionellaceae*?

A) Display a dimorphic lifestyle within a host

B) Are photoautolithotrophs

C) Are obligate intracellular parasites

D) Use metals as terminal electron acceptors

21 INCORRECT

Which microbe is the causative agent of Q fever?

A) *Rickettsia prowazekii*

B) *Legionella pneumophilia*

C) *Coxiella burnetti*

D) *Neisseria gonorrhoeae*

22 INCORRECT

Which of the following is true of the order *Pseudomonadales*?

A) Some members are major plant and animal pathogens

B) Some members are involved in food spoilage

C) Some members are capable of degrading a wide range of organic molecules

D) All of the above

23 INCORRECT

Members of this bacterial genus are able to use a variety of metal electron acceptor.

A) *Pseudomonas*

B) *Thiobacillus*

C) *Shewanella*

D) *Altermonas*

24 CORRECT

Microbes in this order are under investigation as potential use for microbial fuel cells.

A) *Alteromonadales*

B) *Pseudomonadales*

C) *Vibrionales*

D) *Pasteurellales*

25 CORRECT

Which of the following is true of the order *Vibrionales*?

A) Some members are capable of bioluminescence

B) Are chemolithoautotrophs

C) Important pathogens of plants

D) Most are Gram-positive cocci

26 INCORRECT

Which enzyme in *Vibrio* bacteria catalyzes the reaction resulting in bioluminescence?

A) Luminescent

B) Endonuclease

C) Luciferase

D) ATP synthetase

27 INCORRECT

Which of the following best describes the metabolism of bacteria in the order *Enterobacteriales*?

A) Degrade sugars via the Embden-Meyerhoff pathway

B) Ferment glycolytic products

C) Degrade formic acid to hydrogen and carbon dioxide

D) All of the above

28 INCORRECT

Enteric bacteria are divided into groups based on

A) Carbon sources

B) Fermentation products

C) Preferred oxygen regime

D) All of the above

29 INCORRECT

Haemophilus influenza, the causative agent of meningitis, sinusitis, pneumonia, and bronchitis, is a member of which order?

A) *Enterobacteriales*

B) *Vibrionales*

C) *Pseudomonadales*

D) *Pasteurellales*

30 CORRECT

All members of the Delta-proteobacteria are

A) Chemoorganoheterotrophs

B) Chemolithoautotrophs

C) Photolithoautotrophs

D) Photolithoheterotrophs

31 INCORRECT

Which of the following orders are important in the cycling of sulfur in both aquatic and terrestrial ecosystems?

A) *Desulfovibrionales*

B) *Desulfobacterales*

C) *Desulfuromonadales*

D) All of the above

32 CORRECT

Which of the following bacteria exhibit predatory behavior by penetrating the host cell, inhibiting cellular activities, and using available energy sources?

A) *Bdellovibrio*

B) *Shewanella*

C) *Geobacter*

D) *Salmonella*

33 INCORRECT

Which of the following is true of the order *Myxococcales*?

A) Cells move via gliding motility

B) Most members are predators of other microbes

C) Form fruiting bodies that develop myxospores

D) All of the above

34 CORRECT

Social (S) motility exhibits which of the following characteristics?

A) A retractable type IV pili forms at the front end of the cell

B) Occurs in single cells that leave the group

C) Involves slime secretion

D) Involves clusters of cytoplasmic motor proteins (AlgZ)

35 INCORRECT

Infection by which bacterial genus is the most frequent trigger for Guillain-Barre syndrome (GBS)?

A) *Vibrio*

B) *Helicobacter*

C) *Campylobacter*

D) *Pseudomonas*

36 INCORRECT

Members of this genus cause gastritis and peptic ulcer disease.

A) *Vibrio*

B) *Helicobacter*

C) *Campylobacter*

D) *Pseudomonas*

The correct answer for each question is indicated by a ✓.

1 CORRECT

The proteobacteria are the largest and most diverse group of bacteria.

✓ **A)** True

B) False

2 INCORRECT

The most abundant bacteria in the world's oceans are delta-proteobacteria.

A) True

✓ **B)** False

3 CORRECT

Purple non-sulfur bacteria use anoxygenic photosynthesis and possess bacteriochlorophylls *a* or *b*.

✓ **A)** True

B) False

4 CORRECT

Some non-sulfur bacteria are capable of oxidizing very low, non-toxic levels of sulfide to sulfate.

✓ **A)** True

B) False

5 INCORRECT

Rhodospirillum rubrum, a purple non-sulfur bacterium, is only active in oxic conditions.

A) True

✓ **B)** False

6 CORRECT

Members of the genus *Rickettsia* are obligate intracellular parasites.

✓ **A)** True

B) False

7 INCORRECT

Members of the genus *Rickettsia* use glucose as their primary energy source.

A) True

B) False

8 CORRECT

Genome sequencing shows that *Rickettsia prowazekii* is similar in many ways to mitochondria.

A) True

B) False

9 INCORRECT

Members of the families *Caulobacteraceae* and *Hyphomicrobiaceae* reproduce by binary fission.

A) True

B) False

10 INCORRECT

No members of the alpha-proteobacteria exhibit facultative methylotrophy.

A) True

B) False

11 CORRECT

The success of the legume family of plants is based on their association with the *Rhizobiales*.

A) True

B) False

12 INCORRECT

Nitrification makes both nitrate and phosphate available for uptake by plants.

A) True

B) False

13 INCORRECT

The benefits of nitrification are usually long-lasting in an ecosystem.

A) True

B) False

14 CORRECT

Burkholderia cepacia is an environmentally important microbe since it can degrade over 100 different organic molecules.

A) True

B) False

15 CORRECT

Sheaths help bacteria attach to surfaces and protect the cells from predation.

A) True

B) False

16 CORRECT

Members of the order *Hydrogenophilales* exhibit diverse types of metabolism including oxidation of sulfur and denitrification.

A) True

B) False

17 CORRECT

Member of the genus *Thiobacillus* produce sulfuric acid that damages concrete and pipes.

A) True

B) False

18 INCORRECT

Purple sulfur bacteria are usually found in oxygen-rich environments.

A) True

B) False

19 CORRECT

Members of the order *Thiotrichales* form microbial mats at deep hydrothermal ocean vents.

A) True

B) False

20 CORRECT

Thiomargarita namibiensis is the world's largest known bacterium, measuring over 100 micrometers in diameter.

A) True

B) False

21 INCORRECT

Legionella pneumophilia is most infective in the replicative form (RF).

A) True

B) False

22
CORRECT

Coxiella burnetti, the causative agent of Q fever, is an obligate intracellular parasite.

A) True

B) False

23
CORRECT

Pseudomonas aeruginosa infects people with low resistance, such as cystic fibrosis patients.

A) True

B) False

24
CORRECT

Members of the genus *Shewanella* are capable of using metals such as uranium as electron acceptors.

A) True

B) False

25
INCORRECT

Most bacteria are capable of producing nanowires to transport electrons.

A) True

B) False

26
CORRECT

Vibrio cholera exhibits toxin production as a virulence factor.

A) True

B) False

27
CORRECT

Enteric bacteria are capable of glycolysis and fermentation for energy.

A) True

B) False

28
INCORRECT

The orders *Enterobacteriales* and *Pasteurellales* have the nutritional requirements.

A) True

B) False

29
INCORRECT

Sulfate- and sulfur-reducing bacteria (SRB) are important in the cycling of sulfur within the ecosystem.

A) True

B) False

30
INCORRECT

Shewanella is the only bacteria capable of forming nanowires.

A) True

B) False

31
INCORRECT

Predatory bacteria are present in all supergroups of the proteobacteria.

A) True

B) False

32
CORRECT

When *Bdellovibrio*, a predatory microbe, is plated out on agar with host bacteria, plaques will form in the bacterial lawn.

A) True

B) False

33
CORRECT

Most myxobacteria are predators of other microbes.

A) True

B) False

34
CORRECT

The myxobacterial life cycle resembles that of cellular slime molds.

A) True

B) False

35
INCORRECT

Replication of myxobacterial bacteria is dependent on predation.

A) True

B) False

36 INCORRECT

Gliding motility is observed in members of all supergroups of proteobacteria.

A) True

B) False

37 CORRECT

Important human pathogens are members of the epsilon-proteobacteria.

A) True

B) False

The correct answer for each question is indicated by a ✓.

1 INCORRECT

Fungi are members of the domain

A) *Archaea*

B) *Bacteria*

✓ C) *Eukarya*

D) Classification is undetermined

2 INCORRECT

Fungi reproduce

A) Asexually only

B) Sexually only

✓ C) Both sexually and asexually

D) Type of reproduction is undetermined

3 INCORRECT

Which of the following nutritional types best describes the fungi?

A) Heterotrophic

B) Phototrophic

✓ C) Saprophytic

D) None of the above

4 INCORRECT

Fungi are important ecologically because of their ability to

A) Fix atmospheric nitrogen and make it available to other organisms

✓ B) Decompose organic matter to release carbon and nutrients to other organisms

C) Cycle sulfur and iron

D) All of the above

5 INCORRECT

Symbiotic relationships between vascular plant roots and fungi are called

A) Hyphae

B) Mycelium

C) Zooxanthelle

D) Mycorrhizae

6 INCORRECT

Fungi are producers of which of the following industrial products?

A) Breads and cheeses

B) Organic acids

C) Antibiotics and immunosuppressive drugs

D) All of the above

7 CORRECT

The simplest group of fungi are the

A) *Chytridiomycota*

B) *Zygomycota*

C) *Glomeromycota*

D) *Ascomycota*

8 INCORRECT

Members of this group of fungi produce a zoospore with a single, posterior, whiplash flagellum.

A) Ascomycetes

B) Chytrids

C) Basidiomycetes

D) Zygomycetes

9 INCORRECT

The bread mold *Rhizopus* is a member of which fungal group?

A) *Chytridiomycota*

B) *Zygomycota*

C) *Glomeromycota*

D) *Ascomycota*

10 CORRECT

When food becomes scarce or environmental conditions become unfavorable, *Rhizopus* bread mold

A) Reproduces sexually

B) Reproduces asexually

C) Does not reproduce

D) None of the above

11 INCORRECT

Which of the following groups of fungi are mycorrhizal symbionts of vascular plants?

A) *Chytridiomycota*

B) *Zygomycota*

C) *Glomeromycota*

D) *Ascomycota*

12 INCORRECT

Glomeromycetes reproduce

A) Sexually only

B) Asexually only

C) Sexually and asexually

D) Reproduction is undetermined

13 INCORRECT

Which of the following groups of fungi degrade many chemically stable organic compounds including lignin, cellulose, and collagen?

A) *Chytridiomycota*

B) *Zygomycota*

C) *Glomeromycota*

D) *Ascomycota*

14 INCORRECT

A unicellular fungi that reproduces by budding is known as a

A) Mold

Ascus
B)
Yeast
C)
Hyphae
D)

15 INCORRECT

When nutrients are limited, diploid yeast cells undergo

Senescence
A)
Sporulation
B)
Mitosis
C)
Meiosis
D)

16 CORRECT

Filamentous ascomycetes reproduce by producing

Conidia
A)
Sclerotia
B)
Basidia
C)
Asci
D)

17 INCORRECT

Aspergillus mold, the trigger of allergies, asthma, and sinusitis, belongs to the fungal group

Chytridiomycota
A)
Zygomycota
B)
Glomeromycota
C)
Ascomycota
D)

18 INCORRECT

Ergotism, the toxic condition in humans and animals that eat grain infected with fungus, is caused by an organism from the group

Chytridiomycota
A)
Zygomycota
B)
Glomeromycota
C)
Ascomycota
D)

19
CORRECT

Which fungal group includes the stinkhorns, puffballs, toadstools, and mushrooms?

✓ **A)** *Basidiomycota*

B) *Zygomycota*

C) *Glomeromycota*

D) *Ascomycota*

20
INCORRECT

Mushroom toxins damage the host by

A) Attacking liver cells

B) Damaging cells lining the stomach and intestines

C) Inhibiting RNA polymerase

✓ **D)** All of the above

21
INCORRECT

Important plant pathogens causing "rusts" and "smuts" in crop plants are members of which fungal group?

A) *Chytridiomycota*

✓ **B)** *Basidiomycota*

C) *Glomeromycota*

D) *Ascomycota*

22
INCORRECT

Which of the following is true of the Microsporidia?

A) They are obligate endoparasites

B) Lack mitochondria, peroxisomes, and centrioles

C) Have a polar tube for host invasion

✓ **D)** All of the above

Your Results:

The correct answer for each question is indicated by a ✓.

1 INCORRECT

All fungi are members of a monophyletic group.

B) False

2 INCORRECT

Some fungi are prokaryotes.

B) False

3 CORRECT

Fungi can be unicellular or multicellular.

A) True

4 INCORRECT

Fungi are primarily aquatic organisms.

B) False

5 CORRECT

Fungi are saphrophytes and use hydrolytic enzymes to degrade organic material.

A) True

6 INCORRECT

The simplest fungi belong to the group *Zygomycota*.

B) False

7 CORRECT

Chytrids may be responsible for large-scale mortality of amphibians.

A) True

8 INCORRECT

Flagella are absent in all fungi.

B) False

9 CORRECT

A fungus in the genus Rhizopus is the causative agent of seedling blight, a rice disease.

A) True

10 INCORRECT

Rhizopus secretes the toxin responsible for seedling blight rice disease.

B) False

11 INCORRECT

The fungi in mycorrhizal relationships provide carbohydrates to the associated plant.

B) False

12 CORRECT

Glomeromycetes form specialized flat hyphae called appressoria during asexual reproduction.

A) True

13 CORRECT

The causative agents of chestnut blight and Dutch elm disease belong to the *Ascomycota*.

A) True

14 INCORRECT

Yeast cells are capable of reproducing indefinitely.

B) False

15 INCORRECT

Ascospores are released in aquatic environments.

B) False

16 CORRECT

Sclerotia are compact masses of hyphae that can survive in the winter.

A) True

B) False

17 INCORRECT

The fungus responsible for ergotism belongs to the *Zygomycota*.

B) False

18 CORRECT

Exposure to fungal toxins can lead to liver cancer.

A) True

19
CORRECT

Mushrooms used in food production belong to the *Basidiomycota*.

A) True

20
INCORRECT

All mushrooms are edible.

B) False

21
CORRECT

Infection by some members of the Basidiomycota triggers tumor formation in crop plants.

A) True

22
INCORRECT

All fungi are obligate endoparasitic microbes.

B) False

23
INCORRECT

Mitochondria are present in all fungi.

B) False

24
CORRECT

Microsporidia fungi pierce through the host cell membrane with a polar tube.

A) True

The correct answer for each question is indicated by a ✓.

1 INCORRECT

Which of the following is true of the actinobacteria?

A) Source of antibiotics and immunosuppressive drugs

B) Develop filamentous cells

C) Have a complex life cycle

✓ D) All of the above

2 INCORRECT

Actinomycetes sporulation is usually in response to

A) Abundance of phosphate

B) Acidic conditions

✓ C) Nutrient deprivation

D) Increased temperature

3 INCORRECT

Actinobacteria are traditionally classified by

A) Plasma membrane lipids

✓ B) Cell wall sugar content

C) Fermentation products

D) All of the above

4 INCORRECT

Members of the genus *Actinomyces* are responsible for which of the following diseases?

A) Lumpy jaw in cattle

B) Periodontal disease

C) Actinomycoses

✓ D) All of the above

5 INCORRECT

Which of the following members of the *Micrococcineae* can degrade some herbicides and pesticides?

A) *Micrococcus*

B) *Arthrobacter*

C) *Corynebacteria*

D) All of the above

6 INCORRECT

Arthrobacter and *Corynebacteria* reproduce via

A) Sexual reproduction

B) Sporangia

C) Budding

D) Snapping fission

7 INCORRECT

Which of the following actinobacteria have cell walls composed of waxy mycolic acids?

A) *Corynebacterium*

B) *Mycobacterium*

C) *Micrococcus*

D) *Nocardia*

8 INCORRECT

The preferred staining method of identifying mycobacteria is

A) The Gram stain

B) The Wright stain

C) The negative stain

D) The acid-fast stain

9 CORRECT

Mycobacteria cause which of the following diseases in humans?

A) Tuberculosis

B) Acne

C) Ocular infections

D) All of the above

10 CORRECT

Which of the following is true of the *Nocardia*?

A) Degrade hydrocarbons and waxes

B) Cause tuberculosis in humans and cattle

C) Have cell walls composed of mycolic acid

D) All of the above

11 INCORRECT

Which of the following suborders of actinobacteria have an extensive substrate mycelium and type IID cell walls?

A) *Actinomycineae*

B) *Micromonosporineae*

C) *Propionibacterineae*

D) *Streptomycineae*

12 INCORRECT

Members of this actinobacteria suborder are responsible for body odor and acne in humans.

A) *Actinomycineae*

B) *Micromonosporineae*

C) *Propionibacterineae*

D) *Streptomycineae*

13 INCORRECT

Members of this actinobacteria suborder degrade resistant substances in the soil, produce geosmin (the substance responsible for Earth odor), and produce a variety of antibiotics.

A) *Actinomycineae*

B) *Micromonosporineae*

C) *Propionibacterineae*

D) *Streptomycineae*

14 INCORRECT

The first drug used effectively to combat tuberculosis was

A) Penicillin

B) Streptomycin

C) Zithromycin

D) Chloramphenicol

15 INCORRECT

Which of the following actinobacteria suborders have type III cell walls with the sugar derivative madurose?

A) *Actinomycineae*

B) *Streptomycineae*

C) *Frankineae*

D) *Streptosporangineae*

16 INCORRECT

Which of the following actinobacteria suborders have type III cell walls and fix nitrogen to facilitate plant growth?

A) *Actinomycineae*

B) *Streptomycineae*

C) *Frankineae*

D) *Streptosporangineae*

17 INCORRECT

Which of the following actinobacteria genera includes pioneer colonizers of the human intestinal tract, particularly in breastfed babies?

A) *Corynebacterium*

B) *Bifidobacterium*

C) *Mycobacterium*

D) *Nocardia*

1 CORRECT

Actinomycetes are the source of many drugs including antibiotics, immunosuppressives, and antihelminthics.

A) True

B) False

2 INCORRECT

Actinomycetes sporangiospores are extremely heat resistant.

A) True

B) False

3 CORRECT

Actinomycetes are classified primarily based on the structure of their peptidoglycan and cell wall sugar content.

A) True

B) False

4 CORRECT

The suborder *Actinomycineae* are responsible for eye infection and periodontal diseases in human.

A) True

B) False

5 CORRECT

Members of the suborder *Micrococcineae* are usually yellow or red and are inhabitants of mammalian skin and soil.

A) True

B) False

6 INCORRECT

Mycobacteria are effective pathogens because they grow rapidly.

A) True

B) False

7 CORRECT

The waxy, mycolic acid cell wall of the mycobacteria make it extremely resistant to antibiotics.

A) True

B) False

8 CORRECT

The genus *Nocardia* is responsible for biodeterioration of rubber joints in water and sewer pipes.

A) True

B) False

9 CORRECT

Members of the genus *Micromonospora* degrade chitin and cellulose and produce the antibiotic gentamicin.

A) True

B) False

10 INCORRECT

Members of the genus *Streptomyces* are responsible for the production of Swiss cheese and acne in humans.

A) True

B) False

11 CORRECT

Streptomycetes contribute to the smell of moist Earth.

A) True

B) False

12 CORRECT

Streptomycetes are responsible for about 2/3 of antimicrobial agents used in human and veterinary medicine.

A) True

B) False

13 INCORRECT

Most streptomycetes are pathogens.

A) True

B) False

14 CORRECT

Members of the suborder *Frankineae* are associated with root nodules of plants.

A) True

B) False

15 INCORRECT

Bifidobacterium is common in the intestines of bottle fed babies.

B) False

11
INCORRECT

Staphylococci bacteria are usually associated with

A) The intestines

B) The skin

C) The meninges

D) All of the above

12
INCORRECT

Methicillin-resistant *Staphylococcus aureus* (MRSA) most likely developed antibiotic resistance due to

A) Conjugation

B) Mutations

C) Mobile genetic elements

D) Transformation

13
INCORRECT

Members of the genus *Staphylococcus* cause which of the following diseases?

A) Skin infections

B) Food poisoning

C) Toxic shock syndrome

D) All of the above

14
INCORRECT

Which of the following virulence factors are produced by *Staphylococcus aureus*?

A) Biofilm formation

B) Coagulase enzyme

C) Endospores

D) All of the above

15
INCORRECT

Members of this group produce lactic acid as their major or sole fermentation product.

A) *Bacillales*

B) *Lactobacillales*

C) *Listeriaceae*

16
CORRECT

Which of the following is true of the order *Lactobacillales*?

- **A)** Obtain energy via substrate-level phosphorylation
- **B)** They are obligate aerobes
- **C)** Produce ethanol via fermentation
- **D)** Utilize a variety of cytochromes

17
INCORRECT

Which genus is used in the food and dairy industry in the production of Swiss cheese, yogurt, sausage, and sourdough bread?

- **A)** *Clostridia*
- **B)** *Streptococcus*
- **C)** *Lactobacillus*
- **D)** *Listeria*

18
INCORRECT

______________ bacteria are normal residents of the intestinal tracts of humans and most other animals.

- **A)** *Streptococcus pyogenes*
- **B)** *Enterococcus faecalis*
- **C)** *Lactobacillus acidophilus*
- **D)** *Clostridium botulinum*

19
INCORRECT

Streptococcus species are differentiated based on their ability to

- **A)** Develop antibiotic resistance
- **B)** Catabolize amino acids
- **C)** Lyse red blood cells
- **D)** Produce lactic acid via fermentation

20
INCORRECT

Which of the following pathogens is the causative agent of strep throat, glomerulonephritis, and rheumatic fever?

- **A)** *Clostridium tetani*
- **B)** *Streptococcus pyogenes*
- **C)** *Staphylococcus aureus*
- **D)** *Listeria monocytogenes*

1 INCORRECT

Which of the following is the distinguishing feature of the Clostridia?

A) Have compartmentalized cells
B) Are obligate intracellular parasites
C) Strickland reaction
D) Are thermophiles

2 INCORRECT

The bacterial infection responsible for Jim Henson death

A) *Bacillus subtilis*
B) Streptococcus Pyrogenes
C) *Clostridium*
D) Staphylococcus Aureous

3 INCORRECT

Members of the class *Clostridia* are

A) Obligate intracellular parasites
B) Endospore formers
C) Pleomorphic
D) All of the above

4 CORRECT

Members of the genus *Clostridia* are best known for causing

A) Food spoilage
B) Sexually transmitted diseases
C) Anthrax
D) Degradation of petroleum

5 INCORRECT

Members of this species metabolize amino acids since they lack the genes to catabolize sugars.

A) *Clostridium acetobutylicum*
B) *Clostridium tetani*
C) *Clostridium perfringens*
D) All of the above

6 INCORRECT

Members of this genus are anaerobic photosynthetic bacteria that capture light with bacteriochlorophyll *g*.

A) *Clostridium*
B) *Desulfotomaculum*
C) *Heliobacterium*
D) *Veillonella*

7 INCORRECT

Members of this genus metabolize lactate, pyruvate, and malate and are pathogens of the human oral cavity and vagina.

A) *Clostridium*
B) *Desulfotomaculum*
C) *Heliobacterium*
D) *Veillonella*

8 CORRECT

Which of the following species is a model organism for the study of gene regulation and the type species for the order Bacillales?

A) *Bacillus subtilis*
B) *Bacillus cereus*
C) *Bacillus anthracis*
D) *Bacillus sphaericus*

9 INCORRECT

Which of the following is true of members of the genus Bacillus?

A) They produce antibiotics
B) They cause food poisoning
C) They act as insecticides
D) All of the above

10 CORRECT

Which of the following is true of the genus *Thermoactinomyces*?

A) Grow at temperatures between 45 and 60°C
B) Produce toxins
C) Are acidophiles
D) All of the above